Projective Reed–Solomon syndromes beyond redundancy four: deep holes, coherent polar flags, and modular carriers

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Abstract

We classify the projective syndrome directions of deep-hole cosets of $\mathrm{PRS}(q-4)$, of redundancy five, for every prime power $q \geq 7$. Besides the tangent and conjugate-secant families present at lower redundancy, the classification contains two osculating families selected by $q \bmod 3$, a fixed nucleus point and an Artin–Schreier family in characteristic three, and sporadic orbits for $q \in \{7, 8, 9, 11, 13, 17, 19\}$. A syndrome determines a pencil of binary cubics; split-free directions are those whose pencil has no completely split squarefree member. At redundancy five the covering radius is four, so these are precisely the deep-hole directions. The generic pencil is controlled by an off-diagonal fiber-square curve of arithmetic genus one.

At higher redundancy, coherent polar flags replace independent lower-level fibers. They yield a complete all-field deep-hole classification at redundancy six and a complete all-field classification of split-free syndrome directions at redundancy seven. At redundancy seven these directions are deep-hole directions whenever $\rho(\mathrm{PRS}(q-6)) = 6$, in particular for $q \geq 11$. At redundancies eight and nine we prove, respectively for $q \geq 43$ and $q \geq 53$, that the projective deep-hole directions are exactly the persistent tangent and conjugate-secant families. On the latter family the semilinear orbit parameters are the norm-one-torus quotient T/T^{r-1}, modulo inversion and coefficient Frobenius; the tangent orbit decomposition changes when the characteristic divides $r-1$.

In characteristic two the ordinary discriminant degenerates. For the ordered-Hessian pullback we classify the constrained degeneracy locus: its contained carriers are exactly the persistent and Lucas-nucleus carriers. Outside their union we construct an integral genus-at-most-one slice and obtain explicit sufficient field and deletion bounds for a split member. We also determine the Frobenius arithmetic of the power-of-two Lucas family and normalize the first new degree-nine carrier orbit. Its $\mathbb{F}_8$-linear cover is a proper section of an $\mathrm{AGL}_3(\mathbb{F}_2)$ additive cover, and every code-admissible field $\mathbb{F}_{2^m}$ has a split subspace-polynomial witness on this orbit. We do not classify the remaining degree-nine carrier strata or the redundancy-ten deep holes.

1 Introduction

Let $\mathrm{PRS}(k)$ denote the projective Reed–Solomon code of length $q+1$ obtained by evaluating homogeneous binary forms of degree $k-1$ on $\mathbb{P}^1(\mathbb{F}_q)$. If the redundancy is

$$r = q + 1 - k,$$

a parity-check matrix has as columns the $\mathbb{F}_q$-points of the degree-$(r-1)$ normal rational curve

$$\mathcal{R}_{r-1} = \{(1, t, \ldots, t^{r-1}) : t \in \mathbb{F}_q\} \cup \{(0, \ldots, 0, 1)\} \subset \mathbb{P}^{r-1}.$$

Call a projective syndrome direction *split-free* if it is not contained in the span of $r-2$ distinct columns. When $\rho(\mathrm{PRS}(q+1-r)) = r-1$, the split-free directions are precisely the projective syndrome directions of deep-hole cosets. The correspondence between deep holes, one-symbol MDS extensions, and uncovered points of a normal rational curve was developed in [Kai17, ZWK20]. Zhang, Wan, and Kaipa classified projective Reed–Solomon deep-hole directions through redundancy four and identified redundancy five as the next case [ZWK20]. Our first theorem gives the corresponding all-field redundancy-five classification.

Theorem 1.1 (Complete redundancy-five classification)**.** *Let $q \geq 7$ be a prime power. The covering radius of* $\mathrm{PRS}(q-4)$ *is* 4*. Its projective deep-hole syndromes are exactly the following disjoint families:*

(i) *the tangent family* T*, of size* $q(q+1)$*;*

(ii) *the conjugate-secant family* S*, of size* $q(q^2-1)/2$*;*

(iii) *if* $\operatorname{char}\mathbb{F}_q \neq 3$*, the rational osculating-pair family* O^+ *when* $q \equiv 2 \pmod 3$*, of size* $q(q+1)/2$*, or the conjugate osculating-pair family* O^- *when* $q \equiv 1 \pmod 3$*, of size* $q(q-1)/2$*;*

(iv) *in characteristic three, one* $\mathrm{PGL}_2(q)$*-fixed nucleus point and one wild Artin–Schreier orbit* W *of size* $(q^2-1)/2$*;*

(v) *the sporadic* $\mathrm{PGL}_2(q)$*-orbits in Table 1, occurring exactly for* $q \in \{7, 8, 9, 11, 13, 17, 19\}$*.*

There are no other deep-hole syndromes. The three corresponding nonsporadic totals are

$$\frac{q^2(q+3)}{2}, \qquad \frac{q(q+1)(q+2)}{2}, \qquad \frac{q^3+3q^2+q+1}{2},$$

according as $q \equiv 1 \pmod 3$ *with characteristic not three,* $q \equiv 2 \pmod 3$*, or* $\operatorname{char}\mathbb{F}_q = 3$*.*

The main mechanism is a characteristic-free Hankel dictionary. A syndrome at redundancy r annihilates a linear system of binary forms of degree $r-2$. It is split-free if and only if this system contains no completely split squarefree form; it is a deep-hole direction when, in addition, $\rho(\mathrm{PRS}(q+1-r)) = r-1$. At redundancy five the system is a pencil of cubics. Its induced degree-three map of projective lines is either cyclic or has geometric monodromy S_3. Cyclic descent gives the arithmetic families in Theorem 1.1; in the S_3 case the off-diagonal fiber square is an integral curve of arithmetic genus one, and the singular-curve Weil bound [AP95] forces a split fiber for $q \geq 23$. The certified finite computations in Section 9 handle the remaining fields.

Starting at redundancy six, the Hankel system is no longer a pencil. The useful replacement is the coherently parameterized line of contractions obtained by choosing one prospective root. Iteration produces a *polar flag*; every chosen root remains marked because a lower witness containing it would lift with multiplicity two. The pointed coherence is essential: at $q = 19$ an individual contracted fiber has no split witness avoiding its marker, but no consecutive-row polar line is contained in its orbit.

The resulting structural theorem has three outputs.

Theorem 1.2 (Deep-hole and split-free syndrome results beyond five)**.** *The following statements hold.*

(a) *For every prime power $q \geq 7$, the projective deep-hole directions of* $\mathrm{PRS}(q-5)$ *are exactly the persistent tangent and conjugate-secant families, the exceptional orbits listed for $q = 7, 8, 9, 11, 13$, and, when $q = 2^m$ with odd $m \geq 5$, one characteristic-two nucleus orbit.*

(b) *For every prime power $q \geq 7$, the split-free projective syndrome directions of* $\mathrm{PRS}(q-6)$ *are exactly those in the exceptional tables for $q = 7, 8, 9, 11$ and, for $q \geq 13$, the persistent families together with one central nucleus point when $q = 2^m$ and m is odd. Whenever* $\rho(\mathrm{PRS}(q-6)) = 6$ *these are precisely the deep-hole directions; in particular, this holds for $q \geq 11$.*

(c) *For $q \geq 43$, the projective deep-hole directions of* $\mathrm{PRS}(q-7)$*, and for $q \geq 53$ those of* $\mathrm{PRS}(q-8)$*, are exactly the persistent tangent and conjugate-secant families. In either case their total size is $q(q+1)^2/2$.*

At redundancy r, the semilinear orbits in the persistent conjugate-secant family are parameterized by

$$T/T^{r-1} \quad \textit{modulo inversion and Frobenius},$$

where $T = \{z \in \mathbb{F}_{q^2}^{\times} : z^{q+1} = 1\}$ is the norm-one torus and Frobenius is coefficient Frobenius. On the tangent family the unipotent coordinate transforms as $z \mapsto z + (r-1)u$; consequently its orbit decomposition changes precisely when $\operatorname{char} \mathbb{F}_q$ *divides $r - 1$.*

The three parts have different scopes. Part (a) is an all-field deep-hole classification. Part (b) is an all-field classification of split-free syndrome directions and becomes a deep-hole classification exactly when $\rho(\mathrm{PRS}(q-6)) = 6$; this includes every $q \geq 11$. Part (c) is a high-field theorem and makes no classification claim for $q < 43$ at redundancy eight or for $q < 53$ at redundancy nine.

Relation to previous work. The tangent and conjugate-secant families and the coding/normal-rational curve dictionary come from [Kai17, ZWK20]; the covering-radius input in the high-rate range is due to Seroussi and Roth [SR86]. Binary-quartic orbit theory [KPP25], factorization statistics in linear families [CMP17], normal-rational-curve nuclei [GH13], and the general Frobenius/monodromy semantics of splitting families [Wan26] are inputs. Our claim-specific literature searches were refreshed through July 23, 2026, across the pinned coding citation graphs and the named geometric objects, but did not cover MathSciNet. Within that scope, we are not aware of prior proofs of the redundancy-five and redundancy-six projective deep-hole classifications, the redundancy-seven split-free classification, the coherent polar-induction theorem, the high-field persistent-only results at redundancies eight and nine, or the ordered-Hessian and Lucas-carrier results stated here.

Organization. Section 2 gives the Hankel dictionary. Section 3 proves Theorem 1.1. Sections 4–6 develop coherent polar induction and prove Theorem 1.2. Section 7 gives the characteristic-two ordered-Hessian replacement. Section 8 treats Lucas-carrier arithmetic. Section 9 states the computational and formal trust boundary, and Section 10 records the exact open boundary.

2 The syndrome–Hankel dictionary

Let E be a two-dimensional vector space. We use divided powers for syndromes and ordinary symmetric powers for root forms. This avoids invalid binomial rescalings in small characteristic. For $f = (a_0 : \cdots : a_{r-1}) \in \mathbb{P}(\Gamma^{r-1}E)$, write $e_0, \ldots, e_{r-1}$ for the divided-power coordinate basis and $\nu(t) = (1, t, \ldots, t^{r-1})$, with $\nu(\infty) = e_{r-1}$. The perfect divided-power/symmetric-power pairing is characterized by coefficient extraction. Define

$$H_f = \begin{pmatrix} a_0 & a_1 & \cdots & a_{r-2} \\ a_1 & a_2 & \cdots & a_{r-1} \end{pmatrix}, \qquad W_f = \ker H_f \subset \operatorname{Sym}^{r-2} E^{\vee}.$$

Lemma 2.1 (Hankel criterion)**.** *Let $\lambda_1, \dots, \lambda_{r-2}$ be distinct points of $\mathbb{P}(E^\vee)(\mathbb{F}_q)$ and put $g = \prod_i \lambda_i$. Then*

$$f \in \operatorname{span}(\nu(\lambda_1), \dots, \nu(\lambda_{r-2})) \iff g \in W_f.$$

The same statement with repeated factors and Hasse derivatives gives the corresponding osculating spans. Both statements commute with $\mathrm{P\Gamma L}_2(q)$*.*

Proof. On an affine chart, g is the characteristic polynomial of an order-$(r-2)$ recurrence. The length-r solutions are spanned by the geometric sequences $\nu(\lambda_i)$; their independence is the Vandermonde determinant. A root at infinity gives the reversed-chart recurrence. For repeated roots, the confluent Vandermonde basis is formed by Hasse derivatives, so the same argument works in every characteristic. Equivariance follows because substitution transports both the normal rational curve and the recurrence. □

Corollary 2.2 (Split-free criterion)**.** *Assume $\rho(\mathrm{PRS}(q+1-r)) = r-1$. A projective syndrome f is deep if and only if W_f contains no completely split squarefree form of degree $r-2$ over $\mathbb{F}_q$.*

We call the latter property *split-free* even when the covering-radius hypothesis has not been established. Thus every deep syndrome in the radius-$r-1$ range is split-free, while a small-field split-free census is not silently promoted to a code deep-hole classification.

The persistent locus is already visible here. If every member of W_f has a common quadratic factor, then that factor is either a rational square, an irreducible quadratic, or a product of two distinct rational factors. These are respectively the tangent deep family, the conjugate-secant deep family, and a shallow rational secant. The common factor cannot grow with redundancy: since $\dim W_f = r-3$,

$$r-3 \le r-1-\deg\gcd(W_f),$$

so its degree is at most two.

3 Redundancy five: exceptional cubic covers

Here $f = (a_0 : \cdots : a_4)$ and W_f is the pencil annihilated by

$$\begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \end{pmatrix}.$$

The normal rational curve is complete for $q \ge 7$ by the Seroussi–Roth range [SR86], hence the covering radius is four and Corollary 2.2 is unconditional.

3.1 Gcd strata

A quadratic gcd gives exactly the persistent families described above. A linear gcd reduces to a basepoint-free pencil of quadratics. Its degree-two map has a rational deck involution when separable, producing a split member after the finitely many fixed, branch, and collision points are removed; the inseparable characteristic-two case forces f back onto the normal rational curve. The only remaining stratum has trivial gcd.

The incidence of a cubic pencil is a rational curve mapping with degree three to the pencil line. Let $\phi_f : \mathbb{P}^1 \to \mathbb{P}^1$ denote this map. After removing the diagonal from the fiber square, its closure is a bidegree-$(2,2)$ curve $Y_f \subset \mathbb{P}^1 \times \mathbb{P}^1$. A rational point of Y_f away from the diagonal and branch fibers gives two distinct rational roots of one cubic; its third root is then rational.

Lemma 3.1 (Cyclic stratum)**.** *Suppose ϕ_f is geometrically cyclic.*

(i) *In characteristic different from three there are two ramification points. If they are rational, the corresponding osculating-plane intersection is deep exactly when* $q \equiv 2 \pmod 3$*; if they are conjugate, it is deep exactly when* $q \equiv 1 \pmod 3$.

(ii) *In characteristic three all osculating planes meet in* $n = (0:0:1:0:0)$*, whose pencil is* $\langle T^3, U^3 \rangle$*; this fixed point is deep.*

(iii) *The remaining wild characteristic-three pencils have the form* $\langle U^3, T^3 + aTU^2 \rangle$*. They are deep exactly when* $-a$ *is nonsquare, forming one orbit of size* $(q^2-1)/2$.

Proof. In the tame case the deck transformation is rational precisely when Frobenius fixes its multiplier. For rational fixed points this asks $\zeta^q = \zeta$; for a conjugate pair Frobenius also inverts the multiplier. The two conditions are complementary modulo three. In characteristic three the second Hasse derivative of the quartic normal rational curve is constant, giving the nucleus. For the wild normal form, splitting is governed by the additive map $z \mapsto z^3 + az$, which is bijective precisely when its nonzero kernel is not rational, equivalently when $-a$ is nonsquare. The affine stabilizer has order $2q$, giving the orbit size. □

Lemma 3.2 (S_3 stratum)**.** *If* ϕ_f *has geometric monodromy* S_3 *and* $q \geq 23$*, then the pencil contains a completely split squarefree cubic.*

Proof. The S_3 action is transitive on ordered pairs of distinct roots, so Y_f is geometrically integral. Its arithmetic genus is one. The singular-curve Weil bound gives

$$\#Y_f(\mathbb{F}_q) \geq q - 2\sqrt{q}.$$

At most four points lie on the diagonal and at most eight lie over branch values. Since $q - 2\sqrt{q} > 12$ for every prime power $q \geq 23$, a good rational point remains. □

Lemmas 3.1 and 3.2, together with the gcd strata, prove the conceptual part of Theorem 1.1. The exact finite search described in Section 9 closes $7 \leq q \leq 19$ and the isolated field $q = 16$.

Table 1: Complete sporadic orbit inventory at redundancy five. Entries are $\mathrm{PGL}_2(q)$ orbit sizes; exponents denote multiplicity.

q	number of sporadic points	orbit sizes
7	644	$28, 112, 168^3$
8	756	252^3
9	900	$180, 360^2$
11	990	$330, 660$
13	728	$182, 546$
17	1224	1224
19	570	570

The orbit sizes in Table 1 are a compact inventory, not orbit identifiers. The electronic C491 certificate records for each row a canonical normalized representative, stabilizer, complete cubic-pencil member histogram, and Frobenius image; equality with one of those canonical PGL_2 orbits is the finite classification criterion. Thus repeated sizes denote distinct certified orbits.

4 Coherent polar induction

Write $n = r - 1$ and let $f \in \Gamma^n E$. For $\lambda \in E^\vee$, define the divided-power contraction by

$$\langle \iota_\lambda f, g \rangle = \langle f, \lambda g \rangle.$$

Then

$$g \in W_{\iota_\lambda f} \quad \Longleftrightarrow \quad \lambda g \in W_f. \tag{1}$$

The lower witness lifts squarefreely only when $\lambda \nmid g$. Thus the factor selected for contraction must remain as a marked forbidden root.

The first-polar morphism

$$\Phi_f : \mathbb{P}(E^\vee) \longrightarrow \mathbb{P}(\Gamma^{n-1}E), \qquad [\lambda] \longmapsto [\iota_\lambda f]$$

has image a point or line. In coordinates,

$$\Phi_f(r) = (a_1 - ra_0, \ldots, a_n - ra_{n-1}), \qquad \Phi_f(\infty) = (a_0, \ldots, a_{n-1}).$$

Iterating contraction with distinct ordered factors gives the catalecticant polar flag.

Theorem 4.1 (Effective contained-or-transverse induction)**.** *Fix a degree and a number of marked roots. Suppose the lower splitting incidence is stratified so that, off a closed bad carrier, every identity-Frobenius twist is a geometrically integral curve of genus at most g_α, and a divisor of degree at most δ_α removes branch, diagonal, and marker collisions. Suppose further that a noncontained polar line meets the bad carrier in degree at most b and its marked self-collision divisor has degree at most c. Assume scheme-theoretically that if the pulled-back bad carrier or the self-collision divisor is all of $\mathbb{P}^1$, then f lies in the explicitly computed persistent carrier $\mathcal{P}_n$ or modular carrier $\mathcal{M}_n$. An identically zero collision differential is detected and routed to $\mathcal{M}_n$, not counted as a finite divisor.*

Put

$$\mathcal{H}(g, \delta) = \left\lfloor \left(g + \sqrt{g^2 + \delta} \right)^2 \right\rfloor + 1, \qquad Q = \max\{\max_\alpha \mathcal{H}(g_\alpha, \delta_\alpha), b + c\}.$$

For $q \geq Q$, every split-free Hankel system is contained in the explicit persistent catalecticant carrier or an explicit modular-nucleus carrier.

Proof. For a system outside the contained carriers, the bad and self-collision parameters form a divisor of degree at most $b + c$ on $\mathbb{P}^1$. Choose a rational contraction parameter outside it. On the corresponding lower stratum Hasse–Weil gives

$$q + 1 - 2g_\alpha\sqrt{q} > \delta_\alpha,$$

so there is a rational point outside all deletions. It yields a split squarefree lower member avoiding every marker. Equation (1) lifts it to a split squarefree member upstairs, contradicting split-freeness. □

The theorem is conditional only on a finite, level-specific geometric package: equations for the lower bad carrier, geometric integrality of the identity twist, genus/deletion bounds, and contained-line calculations. It is not a claim that these inputs have already been verified in every redundancy.

The containment hypothesis is load-bearing: without it the theorem would prove only escape from an unspecified lower bad carrier. In the PRS applications it is verified by the following two calculations. The contained catalecticant locus is uniform. A consecutive-row identity gives

$$\Phi_f(\mathbb{P}^1) \subset \{\operatorname{rank} C^{(2)} \le 2\} \quad\Longleftrightarrow\quad \operatorname{rank} C_f^{(2)} \le 2.$$

After removing the normal rational curve and rational secants, it gives exactly the tangent and conjugate-secant families. On a sigma fiber the nonsplit torus acts by $z \mapsto z^n$, hence the quotient T/T^n modulo inversion and Frobenius. On a tangent fiber the unipotent cocycle is $z \mapsto z + nu$.

Modular carriers are equally explicit. If $\mathcal{N}$ is a nucleus in the lower divided-power module, the complete space whose first-polar line lies in $\mathcal{N}$ is

$$\mathcal{M}_n(\mathcal{N}) = \mathbb{P}\ker\left(\Gamma^n E \longrightarrow \operatorname{Hom}(E^\vee, \Gamma^{n-1}E/\widetilde{\mathcal{N}})\right).$$

Lucas' theorem computes the nucleus coordinates, and this displayed kernel computes every coherent lift without an ambient syndrome census.

5 Complete classifications at redundancies six and seven

5.1 Redundancy six

The contraction line of a quintic syndrome lies in the redundancy-five syndrome space. On the trivial-gcd stratum, the line meets the secant cubic in degree at most three, the tame cyclic surface in degree at most four, and the pointed ramification divisor in degree at most six. The genus-one lower cover with one marker has deletion degree eighteen; its first prime-power threshold is 29.

In characteristic three the wild carrier is a degree-four cone and no ruling is a polar line. In characteristic two the cyclic carrier degenerates to a plane, whose unique contained polar-line component is

$$\mathcal{N}_3 = \mathbb{P}\langle e_2, e_3\rangle.$$

It is one projective-semilinear orbit of $q+1$ points and is deep exactly over $\mathbb{F}_{2^m}$ with m odd.

The covering-radius gate is complete here. Seroussi–Roth gives $\rho(\mathrm{PRS}(q-5)) = 5$ for $q \ge 9$, and the independent definition scan in the C498 certificate proves the same statement at $q = 7, 8$.

Theorem 5.1 (Redundancy six)**.** *For every prime power $q \ge 7$, the deep syndromes of $\mathrm{PRS}(q-5)$ are the persistent stratum of size $q(q+1)^2/2$, together with:*

(i) *respectively $18, 11, 4, 2, 1$ exceptional $\mathrm{PGL}_2(q)$ orbits at $q = 7, 8, 9, 11, 13$;*

(ii) *one characteristic-two nucleus orbit of size $q+1$ when $q = 2^m$ and odd $m \ge 5$.*

There are no other exceptional orbits. The complete persistent orbit law is T/T^5 modulo inversion and Frobenius, with a tangent fixed/nonzero split exactly in characteristic five.

The five small tables are exhaustive computations. Their semilinear orbit counts are $18, 5, 2, 2, 1$; a collision-energy invariant plus, in odd characteristic, the quintic root-divisor type separates the classes up to precisely Frobenius fusion. The C498 census certificate supplies a canonical representative, stabilizer, member histogram, and Frobenius link for every projective orbit, while the normal-form certificate supplies the collision-energy and root-divisor fingerprint. Orbit counts alone are not used as identifiers.

5.2 Redundancy seven

For a sextic syndrome $f = (a_0, \dots, a_6)$ the contractions are

$$b(r) = (a_1 - ra_0, \dots, a_6 - ra_5).$$

The pointed lower theorem must avoid r. Adding this marker raises the genus-one deletion budget to 24, whose first prime-power threshold is 37. A noncontained polar line meets the lower secant closure at most three times; marked ramification of the moving g^3_5 has degree at most eight.

The characteristic-two nucleus line at redundancy six has exactly one coherent lift, the central point e_3, with

$$W_{e_3} = \langle 1, t, t^4, t^5 \rangle.$$

It is deep for odd m; for even m, the homogenization of $t^4 + t$ has the five roots $\mathbb{P}^1(\mathbb{F}_4)$.

Theorem 5.2 (Redundancy-seven split-free classification)**.** *For every prime power $q \geq 13$, the split-free syndromes at redundancy seven are the persistent stratum, together with the central nucleus point when $q = 2^m$ and m is odd. At $q = 7, 8, 9, 11$ the additional split-free orbit-size profiles are:*

7	$56,\ 84^5,\ 112^2,\ 168^{45},\ 336^{141}$
8	$63,\ 72,\ 84^3,\ 168^4,\ 252^{24},\ 504^{86}$
9	$180^3,\ 240^6,\ 360^{18},\ 720^{27}$
11	$264^2,\ 440,\ 660^2.$

This is the complete all-field split-free classification. Since Seroussi–Roth gives $\rho(\mathrm{PRS}(q-6)) = 6$ for every prime power $q \geq 11$, it is also the complete deep-hole classification in that range. At $q = 7, 8, 9$ the table is not promoted to a code deep-hole classification without a separate covering-radius calculation. The persistent orbit law is T/T^6 modulo inversion and Frobenius, with tangent splitting in characteristics two and three.

For the four finite rows the C509 certificate contains the canonical representative, stabilizer, persistent/central flags, and Frobenius link of every orbit. The displayed size profile is only a compact summary; repeated sizes are distinguished by those canonical records.

The $q = 19$ lower net $\langle 1, t^3, t^4 \rangle$ has six split members, all containing the marked point at infinity, yet no coherent sextic polar line lies in its orbit. This is the concrete reason Theorem 4.1 is formulated for parameterized flags rather than a set of bad lower fibers.

6 Fixed-level high-field theorems

6.1 Redundancy eight

After three contractions the bottom object remains the same geometric-S_3 ordered-pair curve used at redundancy five. The diagonal and branch deletion costs twelve and three marked roots cost eighteen, so $\delta = 30$. The exact inequality

$$q + 1 - 2\sqrt{q} > 30$$

holds at every prime power $q \geq 43$. The first-polar line has transverse/collision budget

$$3 + 1 + 10 = 14,$$

where ten is the ramification degree of a moving g^4_6.

The complete nucleus-lift calculation leaves characteristic-three and characteristic-five lines, both shallow in this range; the characteristic-two central component has zero further lift.

Theorem 6.1 (Redundancy eight). *For every prime power $q \geq 43$, the deep syndromes of* $\mathrm{PRS}(q-7)$ *are exactly the persistent tangent and conjugate-secant families, of total size* $q(q+1)^2/2$. *Their orbit law is* T/T^7 *modulo inversion and Frobenius. The tangent family splits exactly in characteristic seven.*

This theorem makes no assertion about additional deep orbits for $q < 43$.

6.2 Redundancy nine

Four markers give deletion degree 36 and the first prime-power solution of $q + 1 - 2\sqrt{q} > 36$ is 53. The transverse/collision budget is at most $3 + 2 + 12 = 17$.

The only substantial modular carrier occurs in characteristic seven and is a projective binary-quartic space. Fixing a split quintic $P(t) = \sum_{i=0}^{5} p_i t^i$ and a carrier point $h = (a_0, \dots, a_4)$, put

$$H_j = \sum_{i=0}^{4} a_i p_{i+j}$$

with $p_k = 0$ outside $0 \leq k \leq 5$, and and

$$D = H_0 H_2 - H_1^2, \quad N_s = H_{-1} H_2 - H_0 H_1, \quad N_u = H_{-1} H_1 - H_0^2.$$

Off $D = 0$, the unique residual quadratic is

$$Dt^2 - N_s t + N_u,$$

with branch polynomial $K = N_s^2 - 4 N_u D$. Every squarefree geometric binary quartic is transported to the standard one-parameter normal form used in the slice calculation. For six choices of four fixed roots, the six slice discriminants have unit gcd in that parameter, so at least one transported slice is geometrically integral. Every multiple-root normal form likewise has an integral reduced slice of genus at most one. Thus there is no exceptional geometric quartic orbit.

Theorem 6.2 (Redundancy nine). *For every prime power $q \geq 53$, the deep syndromes of* $\mathrm{PRS}(q-8)$ *are exactly the persistent tangent and conjugate-secant families, of total size* $q(q+1)^2/2$. *Their orbit law is* T/T^8 *modulo inversion and Frobenius; the tangent family splits exactly in characteristic two.*

On the formal characteristic-seven binary-quartic carrier at $q = 7$, precisely 819 *projective quartics are rootless and split-free; this is a carrier calibration, not a deep-hole statement for the inadmissible parameter* $\mathrm{PRS}(-1)$. *Every carrier point is shallow for the actual code parameters at $q = 49$ and at every $q = 7^m \geq 343$.*

The $q = 49$ clause is an exhaustive computation on the 5,884,901 carrier points, not an ambient $\mathbb{P}^8$ census. The $q \geq 343$ clause is conceptual: base selection plus the integral genus-at-most-one residual cover gives a split witness.

7 Characteristic two: the ordered Hessian

In characteristic two the divided binary-cubic discriminant becomes the doubled quadric $(AD + BC)^2$, so its pullback cannot distinguish generic splitting. The correct replacement retains an ordered residual root.

Let $\ell = \mathbb{P}\langle U, V\rangle \subset \mathbb{P}(\Gamma^3 E)$ and write

$$\mathrm{Hess}^{\mathrm{div}}(uU + vV) = N_u(u,v) X^2 + N_s(u,v) XY + D(u,v) Y^2.$$

The ordered-Hessian incidence

$$N_u(u,v)X^2 + N_s(u,v)XY + D(u,v)Y^2 = 0 \tag{2}$$

has bidegree $(2,2)$ and arithmetic genus one. Off $N_s = 0$, its Artin–Schreier class is DN_u/N_s^2.

The parameter space of cubic lines is $G(1, \Gamma^3 E)$. Let

$$\Sigma = \{\mathbb{P}(QE) : [Q] \in \mathbb{P}(\mathrm{Sym}^2 E)\} \subset G(1, \Gamma^3 E)$$

be the Veronese surface of common-quadratic lines. In the ambient cubic space, let $\mathcal{Q}_0 = \{AD + BC = 0\} \subset \mathbb{P}(\Gamma^3 E)$. Its two families of lines give two Fano ruling conics in the Grassmannian; one is the tangent-conic boundary of Σ, and we call the other $\mathcal{N}$. A root-compatible pullback is the family of lines obtained by fixing a split factor R and varying the final two linear factors in the consecutive Hankel contraction of a syndrome.

Theorem 7.1 (Ordered-Hessian component and containment theorem)**.** *After removing the forced vertical factors from intersections with the twisted cubic, the complete geometric degeneracy locus of* (2) *consists of:*

(i) *the Veronese surface* Σ *of lines* $\mathbb{P}(QE)$*, on which the incidence is separably reducible; and*

(ii) *the two Fano ruling conics of* $\mathcal{Q}_0$*, on which the root projection of* (2) *is inseparable.*

The complementary ruling $\mathcal{N}$ *has no nontrivial rank-two root-compatible Hankel pullback. Hence the complete contained pullback is exactly the persistent catalecticant and Lucas-nucleus carrier union.*

For syndrome degree $n \geq 5$*, outside that union an integral reduced slice exists whenever*

$$q \geq \min\left\{\frac{(n-4)(n+3)}{2} + 1,\ 5(n-4)\right\}.$$

Its deletion degree is at most $3n - 4$*. If also* $q + 1 - 2\sqrt{q} > 3n - 4$*, every split-free syndrome lies in the persistent/Lucas carrier union.*

Proof outline. Once vertical and horizontal factors are removed, any separable factorization of a $(2,2)$ curve must be $(1,1) + (1,1)$. The two factors are graphs of projectivities. Semisimple and unipotent normal forms show that only the lines QE occur. The ambient equation $AD+BC = 0$ gives two quadric rulings. Substituting the consecutive Hankel contraction forms into the complementary ruling forces overlapping triples of linear forms to be proportional, so the root-compatible line has rank at most one.

For a fixed split base R of degree $n-4$, the Plücker coordinates of the root-compatible line are quadratic in the coefficients of R. The degeneracy equations have degree at most four in each root. Schwartz–Zippel with the Vandermonde gives the quadratic bound; using pairwise disjoint five-element interpolation blocks gives $q \geq 5(n-4)$. After a good base is selected, the five deletion divisors have total degree $3n - 4$, and Hasse–Weil supplies a good rational point. $\square$

Theorem 7.1 is a containment theorem. It does not assert that every Lucas-carrier point is deep; that remains an arithmetic question for each level.

8 Lucas carriers and their arithmetic

Let $d = 2^s$ with $s \geq 2$ in characteristic two. Lucas' theorem gives the top nucleus of the degree-d normal rational curve as

$$\mathbb{P}\langle e_1, \dots, e_{d-1}\rangle,$$

whose coherent degree-$(d+1)$ lift is

$$\mathcal{M}_{d+1} = \mathbb{P}\langle e_2, \dots, e_{d-1}\rangle.$$

At the distinguished Borel endpoint e_{d-1} the Hankel kernel contains

$$\mathcal{U}_s = \langle 1, t, t^d\rangle.$$

Proposition 8.1 (Linearized root cover)**.** *On the generic locus $BC \neq 0$, the normalized ordered-root cover of $A + Bt + Ct^d$ has geometric monodromy $\mathrm{AGL}_1(\mathbb{F}_d)$, minimal constant field $\mathbb{F}_d$, and based deck group C_{d-1}. Coefficientwise Frobenius acts on the based group by multiplication by two, with exact order s. The net contains a split squarefree member over $\mathbb{F}_{2^m}$ if and only if $s \mid m$.*

Proof. After dividing by C, choose

$$\beta^{d-1} = B/C, \qquad y^d + y = A/(C\beta^d).$$

The roots are $\{\beta(y+c) : c \in \mathbb{F}_d\}$. Their affine symmetries give $\mathrm{AGL}_1(\mathbb{F}_d)$, and ratios of root differences recover the constant field. A split root set forces $\mathbb{F}_d \subset \mathbb{F}_{2^m}$; conversely $t^d + t$ splits simply whenever this inclusion holds. $\square$

At $s = 2$ this recovers the odd/even extension law at redundancies six and seven. The first fresh carrier is $s = 3$, in syndrome degree nine. The $\mathbb{F}_8$-linear section has order-three component Frobenius, but its four quotient directions change the full answer.

Theorem 8.2 (The degree-nine e_7 orbit)**.** *At the distinguished point,*

$$W_{e_7} = \langle 1, t, t^2, t^3, t^4, t^5, t^8\rangle.$$

After ordering roots $0, 1, a, b, c, d, u, v$ and putting

$$S = a+b+c+d, \quad E = ab+ac+ad+bc+bd+cd, \quad h = 1+S, \quad p = 1+E+S+S^2,$$

work on the open set where a, b, c, d are pairwise distinct and avoid $0, 1$, where $h \neq 0$, and where the remaining collision resultants do not vanish. With $y = u/h$, the remaining pair is the geometrically integral Artin–Schreier cover

$$y^2 + y = p/h^2.$$

Its rational lifting law is $\mathrm{Tr}_{\mathbb{F}_{2^m}/\mathbb{F}_2}(p/h^2) = 0$.

The directions t^2, t^4 contain the additive subcover

$$t^8 + A_4t^4 + A_2t^2 + A_1t + A_0, \qquad A_1 \neq 0,$$

whose affine-frame monodromy is $\mathrm{AGL}_3(\mathbb{F}_2)$. For every $m \geq 3$, every three-dimensional $\mathbb{F}_2$-subspace $V \subset \mathbb{F}_{2^m}$ gives the split squarefree subspace polynomial

$$L_V(t) = \prod_{v \in V}(t - v) = t^8 + A_4t^4 + A_2t^2 + A_1t.$$

Consequently the entire PGL_2-orbit of e_7 is shallow over every $\mathbb{F}_{2^m}$ with $m \geq 3$ (and hence over every admissible redundancy-ten field). The number of monic additive-affine witnesses is

$$\frac{q(q-1)(q-2)(q-4)}{1344}.$$

Thus the order-three law of Proposition 8.1 belongs to a proper $\mathbb{F}_8$-linear section, not to deepness of the full e_7 kernel. The other intrinsic strata of the six-dimensional degree-nine carrier are not classified here.

9 Verification and formal boundary

The conceptual theorems above are accompanied by deterministic, committed evidence bundles. Table 2 records which claims use exhaustive computation. Every named bundle contains the primary generator, a compact JSON or text certificate, an independent replay where stated, and a SHA-256 manifest. The load-bearing generator, certificate, and replay hashes were rechecked during manuscript integration. The C491 and C498 manifests also include their living theorem reports; those two report-only hashes became stale after later closeout expansions. No computational artifact hash is stale, and no predecessor manifest is silently rewritten here.

Table 2: Paper-facing verification map. The paths are relative to the repository's `notes/` directory.

Result	Generator/certificate	Independent boundary
R5	`2026-07-22-c491-prs-deep-hole-census.py`, `2026-07-22-c491-prs-deep-hole-census.json`	`2026-07-22-c491-prs-deep-hole-replay.py`; `2026-07-22-c491-prs-redundancy-five.sha256`; exhaustive on nineteen fields, pointwise and orbitwise
R6	`2026-07-22-c498-prs-deep-hole-census.rs`, `2026-07-22-c498-prs-deep-hole-census.json`	`2026-07-22-c498-prs-deep-hole-replay.py`; `2026-07-22-c498-prs-redundancy-six.sha256`; independent direct scan through $q = 16$, structural replay thereafter
R6 small	`2026-07-23-c498-small-exceptional-normal-forms.py`, `2026-07-23-c498-small-exceptional-normal-forms.json`	same-file deterministic checker; no second implementation; `2026-07-23-c498-small-exceptional-normal-forms.sha256`
R7	`2026-07-23-c509-prs-deep-hole-calibration.py`, `2026-07-23-c509-prs-deep-hole-calibration.json`	`2026-07-23-c509-prs-deep-hole-calibration-replay.py`; independent five-secant and orbit checks; `2026-07-23-c509-prs-redundancy-seven.sha256`
R8	`2026-07-23-c513-prs-redundancy-eight.py`, `2026-07-23-c513-prs-redundancy-eight.json`	`2026-07-23-c513-prs-redundancy-eight-replay.py`; algebra, nuclei, bounds, not an ambient census; `2026-07-23-c513-prs-redundancy-eight.sha256`
R9	`2026-07-23-c516-prs-redundancy-nine.py`, `2026-07-23-c516-prs-redundancy-nine.json`	`2026-07-23-c516-prs-redundancy-nine-replay.py`; independent residual algebra and slice checks; `2026-07-23-c516-prs-redundancy-nine.sha256`
R9 $q = 49$	`2026-07-23-c516-prs-redundancy-nine-q49.rs`, `2026-07-23-c516-prs-redundancy-nine-q49.txt`	one exhaustive carrier implementation; residual formula independently replayed; C516 manifest above
Hessian	`2026-07-23-c525-ordered-hessian-arf-pullback.py`, `2026-07-23-c525-ordered-hessian-arf-pullback.json`	`2026-07-23-c525-ordered-hessian-arf-pullback-replay.py`; bounded algebra regression, not the geometric proof; `2026-07-23-c525-ordered-hessian-arf-pullback.sha256`
Lucas	`2026-07-23-c529-characteristic-two-lucas-carrier-arithmetic.py`, `2026-07-23-c529-characteristic-two-lucas-carrier-arithmetic.json`	`2026-07-23-c529-characteristic-two-lucas-carrier-arithmetic-replay.py`; `2026-07-23-c529-characteristic-two-lucas-carrier-arithmetic.sha256`
e_7	`2026-07-23-c530-degree-nine-lucas-e7-quotient-cover.py`, `2026-07-23-c530-degree-nine-lucas-e7-quotient-cover.json`	`2026-07-23-c530-degree-nine-lucas-e7-quotient-cover-replay.py`; `2026-07-23-c530-degree-nine-lucas-e7-quotient-cover.sha256`

The redundancy-nine residual algebra and synthesis implication are also formalized in Lean. The import gate and its axiom-audit module are

`RelativeConicArcs.Gates.PRSRedundancyNine`,

`RelativeConicArcs.Gates.PRSRedundancyNineAxiomAudit`.

Lean checks divided-power contraction, both residual Hankel identities, the determinant and branch formulas, the residual roots and their sum/product, the witness-to-shallow implication, the persistent-family cardinality, and the orbit-count table from explicit arithmetic-case and family-count inputs. It does not derive that table from a formal $\mathrm{PGL}_2 \,/\, \mathrm{P\Gamma L}_2$ action. Geometric integrality, Hasse–Weil existence after deletions, identification of geometric families with PRS syndromes, and exhaustion remain explicit structure fields rather than hidden axioms. The audit reports only the standard `propext`, `Classical.choice`, and `Quot.sound` dependencies.

10 Exact boundary and further questions

The results establish a paper spine, not an arbitrary-redundancy classification.

1. Redundancy five is complete for every $q \geq 7$.
2. Redundancy six is a complete all-field deep-hole classification. Redundancy seven is a complete all-field split-free syndrome classification and a deep-hole classification for $q \geq 11$; its $q = 7, 8, 9$ radius gate remains separate.
3. Redundancies eight and nine are classified only in the displayed high-field ranges. No bounded-field completion is inferred.
4. The characteristic-two ordered-Hessian theorem classifies contained carriers; it does not classify arithmetic deepness on every Lucas carrier.
5. The e_7 orbit is only one intrinsic stratum of the degree-nine Lucas carrier. The other strata, and therefore a redundancy-ten synthesis, remain open.
6. The effective polar-induction theorem requires a new level-specific monodromy/component calculation at each redundancy. A field census cannot replace this input.

The structural conclusion is nevertheless stable. Beyond redundancy four, deepness is no longer controlled by one invariant or by isolated bad fibers. The persistent geometry is a coherent rank-two polar flag; the modular geometry is a Lucas-computed contraction kernel; and the remaining arithmetic is the Frobenius action on a normalized ordered-root cover. Redundancy five is the first exceptional-cover case, while the later theorems show how that cover is propagated, contained, or replaced in the levels and carrier strata treated here.

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